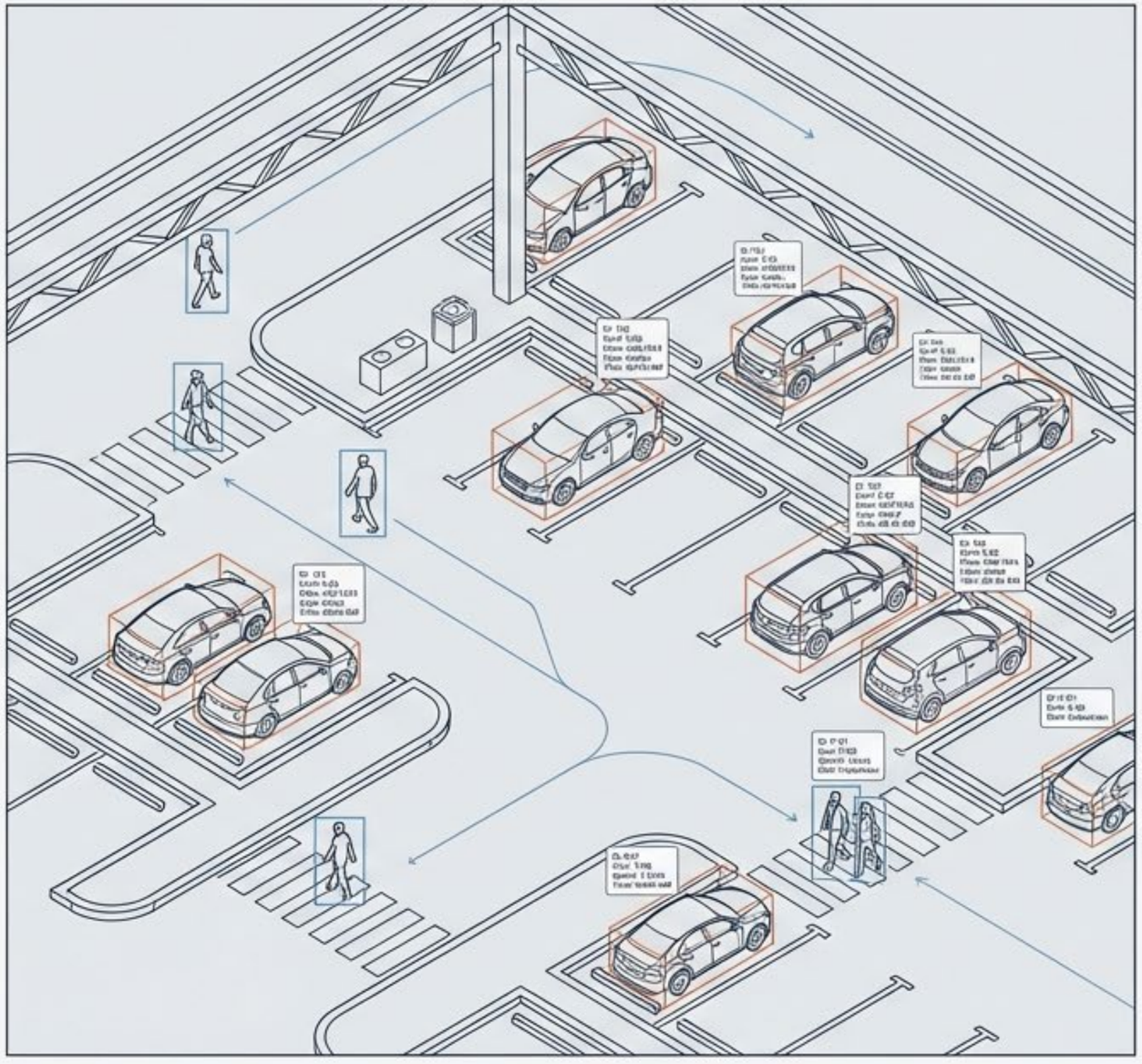
Ctrl+Park: Turning Raw Angled CCTV into Real-Time Parking Intelligence

A software-only computer vision solution requiring no dedicated top-down sensors or proprietary hardware.



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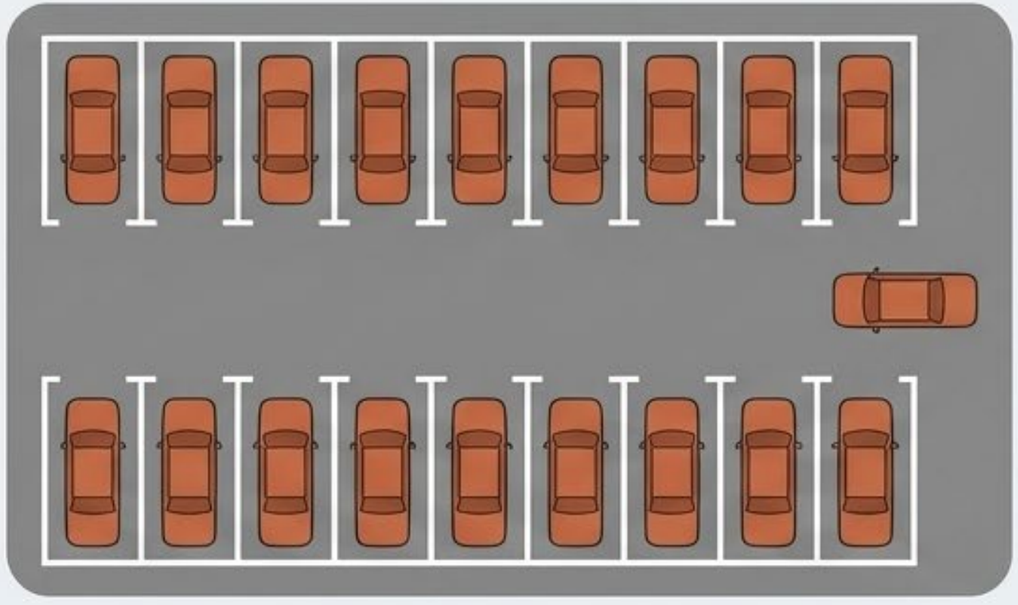
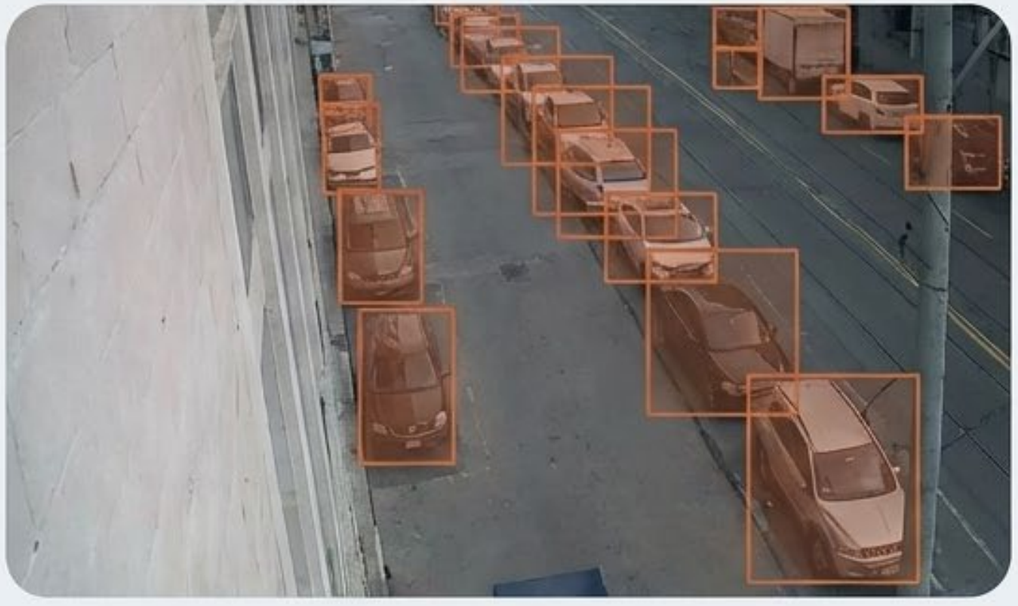
The Real-World Data Gap: Why Angled CCTV?

Theoretical: Top-Down Cameras



- Well-studied domain with simple geometry
- Open-source solutions readily available
- **The Problem**: Exceedingly rare in actual urban The Problem: Exceedingly rare in actual urban deployments due to high installation costs.

Practical: Angled CCTV



- The dominant physical installation worldwide
- High perspective distortion and dynamic scale variation
- **The Solution**: Ctrl+Park is engineered specifically for The Solution: Ctrl+Park is engineered specifically for this hard domain, unlocking value from existing cameras.

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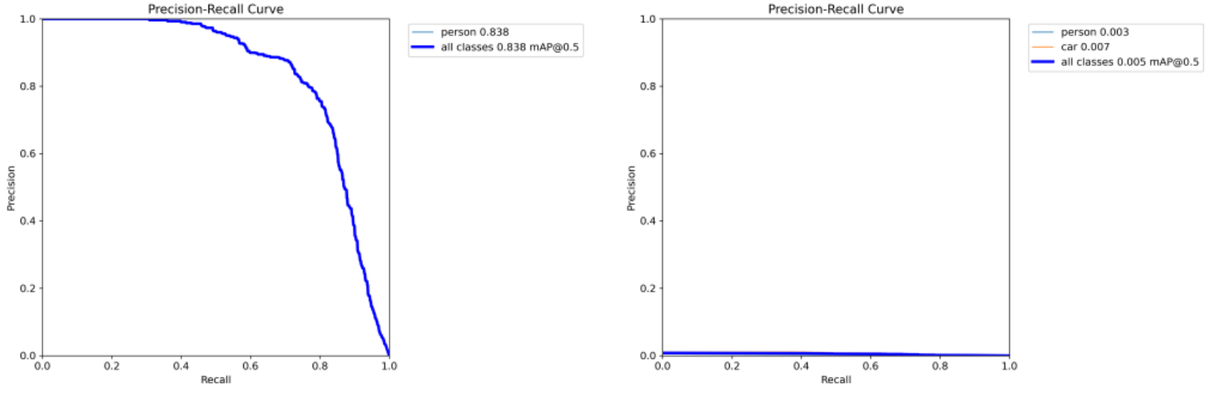
The Baseline Failure & The Domain Leap

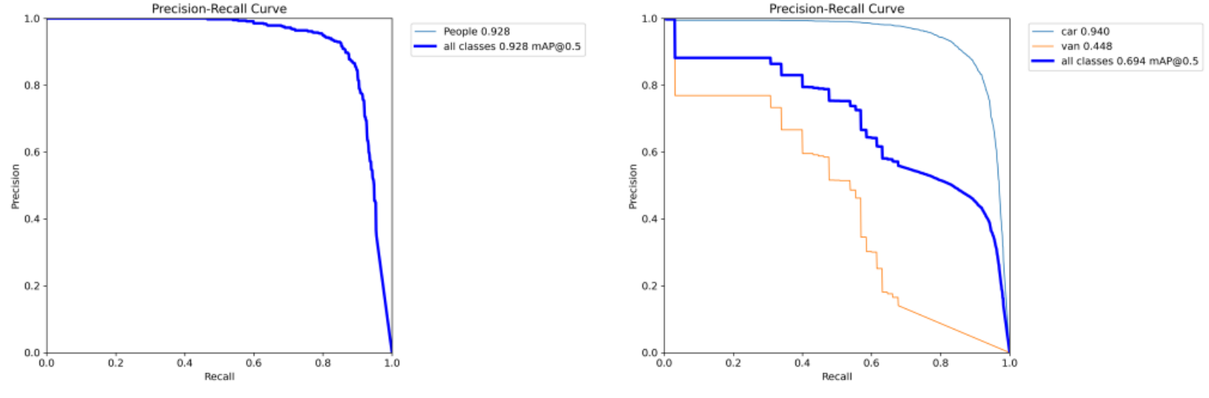
Generic AI fails in parking lots. Domain-specific fine-tuning is an absolute necessity.

Metric	Baseline (COCO)	Ctrl+Park (Fine-Tuned)
Car Detection (mAP@50)	0.7%	**94.0%**
Pedestrian Detection (mAP@50)	83.8%	92.8%
Performance Multiplier	-	**128x Leap**

Baseline Model (left: people, right: cars)



Finetuned Models (left: people, right: cars)



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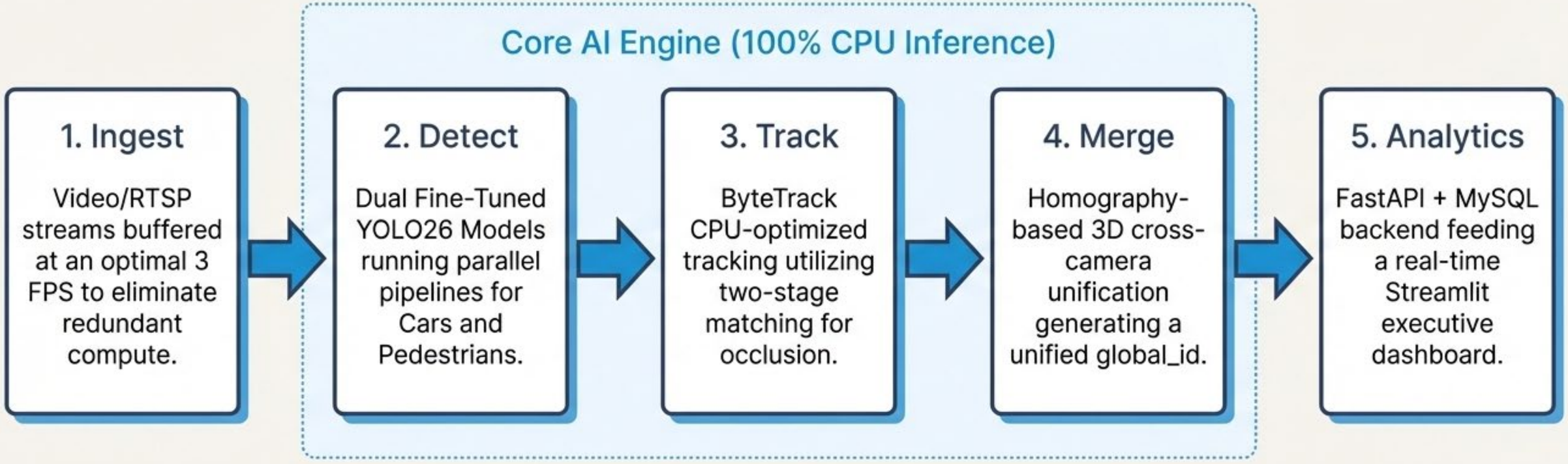
The Ctrl+Park Engine: End-to-End Pipeline Architecture

A scalable, hardware-agnostic data pipeline transforming video into structured SQL data.



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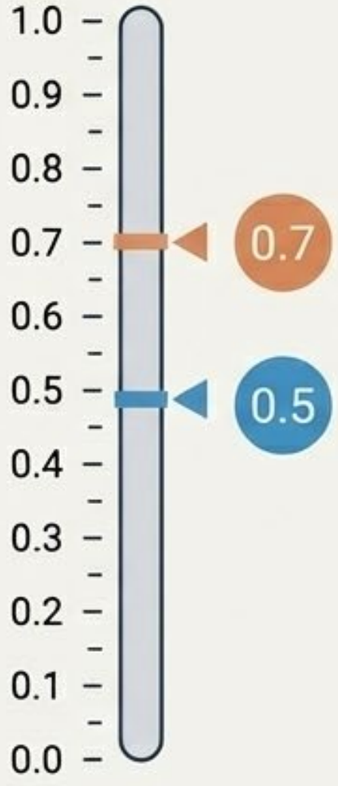
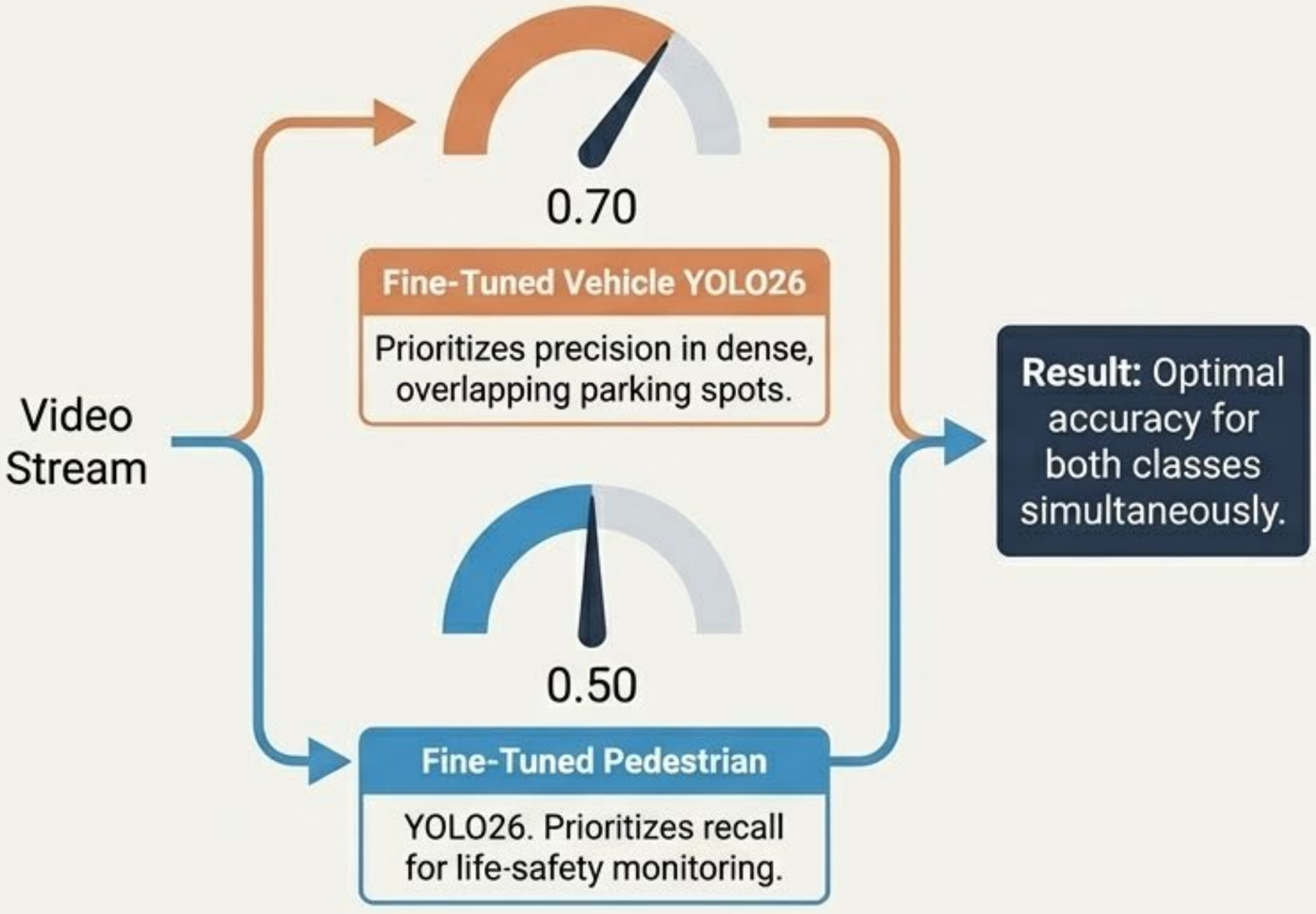
Innovation I: The Threshold Dilemma & Dual-Model Detection



High Threshold: Separates dense parked cars, but misses pedestrians entirely.

Low Threshold: Captures pedestrians, but creates ghost duplicate bounding boxes on cars.

A single multi-class model forces an unacceptable compromise.



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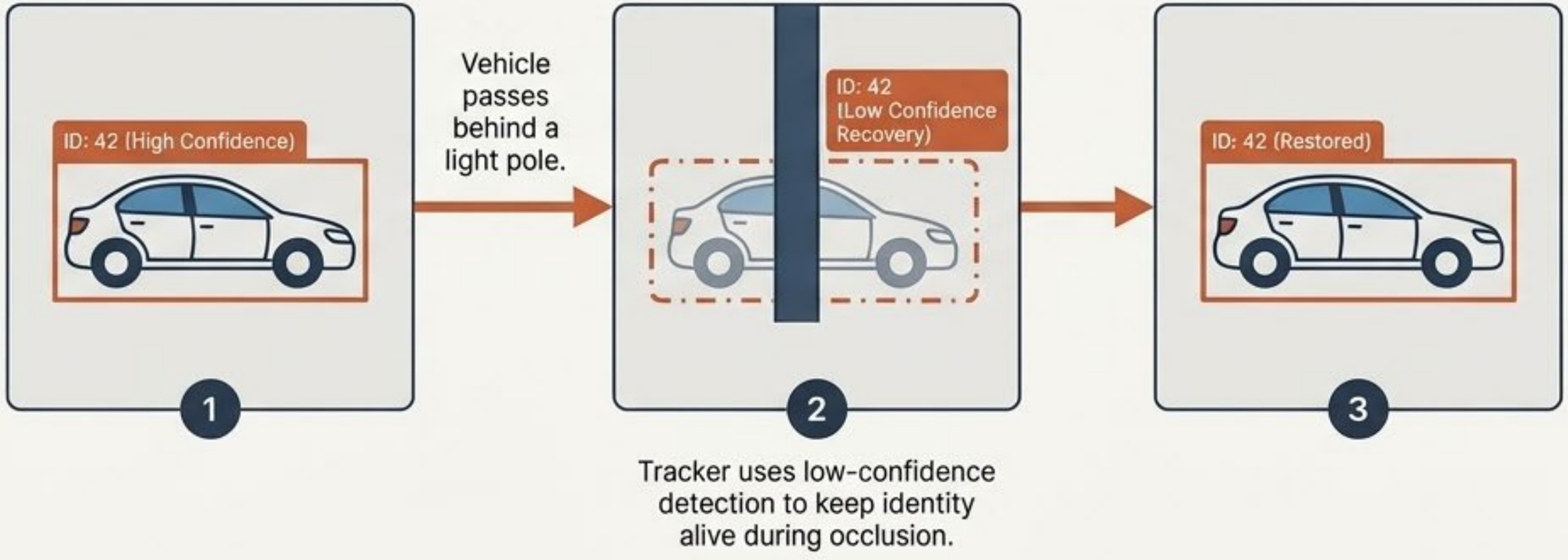
Innovation II: CPU-Optimized Tracking & Occlusion Recovery



The Business Value

- Eliminates heavy appearance re-identification models (like DeepSORT).
- Achieves **150-200 ms**/frame inference purely on a standard CPU.
- **Zero GPU Requirement** drastically lowers operational deployment costs.

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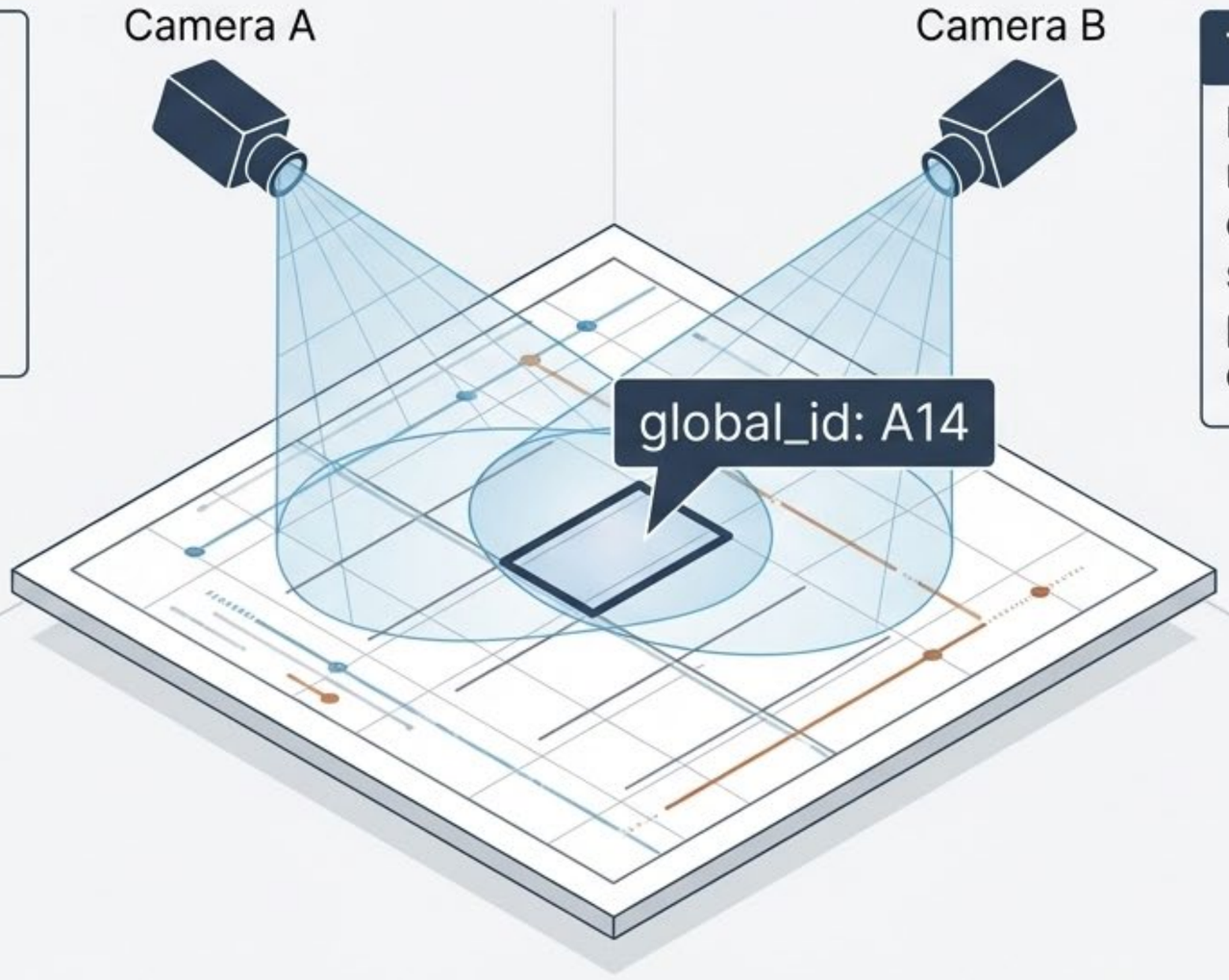
Innovation III: Cross-Camera 3D Unification

Solving the double-counting problem by mapping 2D pixels to a unified physical reality

The Problem:

A single physical vehicle seen by two overlapping cameras generates generates conflicting data points.



The Solution:

Homography calibration maps independent 2D camera feeds onto a single metric coordinate plane, creating a true digital twin.

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From Raw Video to Structured Analytics

Every metric extracted automatically without manual labeling or ground loop sensors.

Ctrl+Park Streamlit

Overview

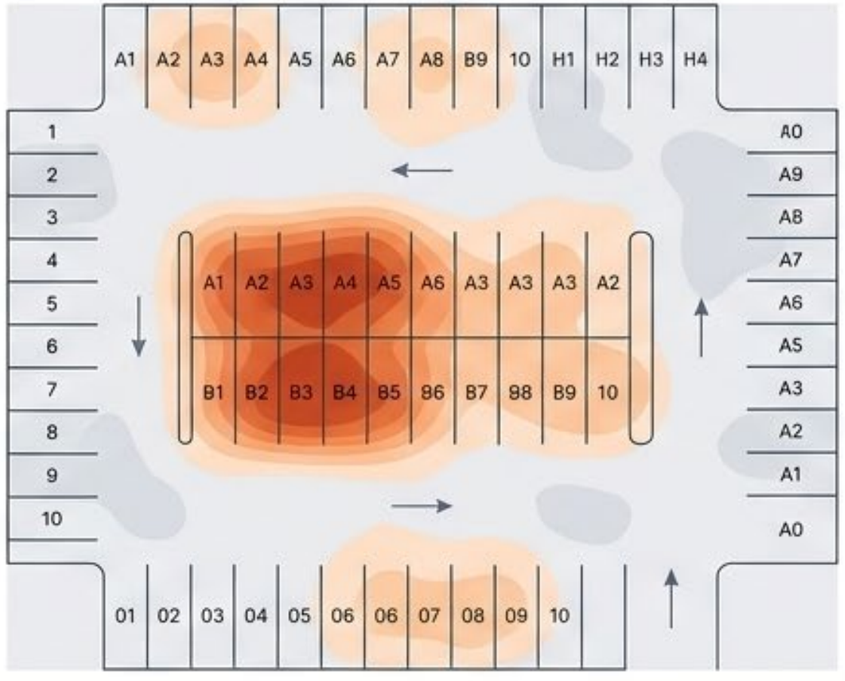
Tracking

3D Simulation

Reports

KPI

Total Tracked:	Current Occupancy:	Avg Dwell Time:	Exits Today:
1,402	84%	2h 15m	342



Live Event Log

Time	Event	Details
10:45:12 AM	Slot A12	Occupied (Dwell: 42m) - Vehicle Detected (Orange Alert Icon)
10:45:10 AM	Slot B04	Free - Slot Cleared
10:44:55 AM	Slot B05	Occupied (Dwell: 12m) - Vehicle Detected (Orange Alert Icon)
10:44:30 AM	Pedestrian Alert	Aisle 3 - Person Detected
10:44:12 AM	Slot A01	Aisle 3 - Person Detected
10:44:10 AM	Slot B04	Free - Slot Cleared
10:44:55 AM	Slot B05	Free - Slot Cleared
10:44:55 AM	Slot B04	Aisle 3 - Person Detected
10:44:30 AM	Pedestrian Alert	Aisle 3 - Person Detected

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The Business ROI: Monetizing Parking Intelligence

Transforming structured occupancy data into direct commercial value for operators.

Dynamic Pricing Engine

Automatically raise parking tariffs during peak demand by linking real-time occupancy data to a dynamic pricing API.

+15-30% Revenue Uplift

Revenue Leakage Detection

Compare physical dwell time captured by the cameras against the paid ticket duration to flag violators instantly.

Recovers 5-12% Lost Revenue

ESG & Compliance Reporting

Quantify precise CO2 emissions generated by vehicles circling to find open spots, fulfilling municipal sustainability mandates.

Automated ESG Metrics

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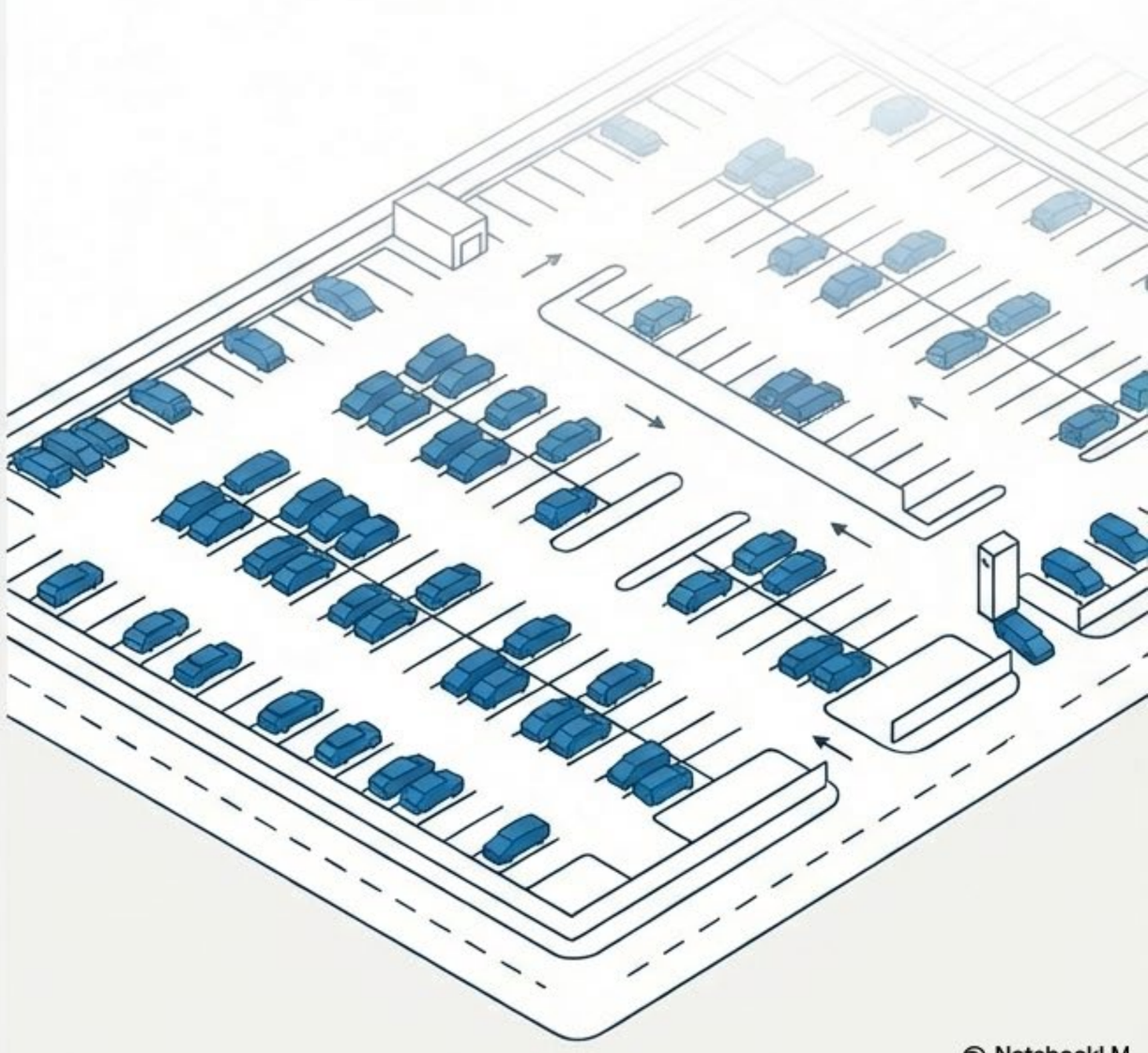
Ctrl+Park: Ready for Deployment

- **94.0% mAP** on real-world, angled footage.
- **Hardware agnostic**: runs on existing CCTV.
- **100% CPU inference** ensures zero GPU costs.
- **Multi-camera 3D** unified tracking.

The cameras are already recording.
The data is already there.
Ctrl+Park turns it into **value.**



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